# Insup Lee

Cybersecurity & AI Researcher

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## Research Interests

My primary research focuses on **AI for cybersecurity and threat intelligence**. I have developed **domain-specific generative models** that preserve structural properties downstream classifiers require, ranging from continuous physical signals in wireless communications to discrete TTP sequences in cyber threat analysis. I recently expanded this foundation to LLMs for cyber threat knowledge graph extraction, shifting my focus from synthetic data generation to automated intelligence extraction. To advance this trajectory, I intend to deploy **agentic AI systems** to automate complex security tasks and investigate their vulnerabilities, contributing to the field of **securing AI**.

## Education

**Korea University** — Seoul, Korea
*Ph.D. in Cybersecurity* — Sep 2019 - Present
- Advisors: Sangjin Lee and Seokhie Hong
- Dissertation: Domain-Specific Generative Models for Data Augmentation in Cybersecurity

**Korea University** — Seoul, Korea
*B.E. in Cyber Defense* — Mar 2014 - Feb 2018

## Work Experience

**Korea University** — Seoul, Korea
*Lecturer* — Sep 2025 - Feb 2026
- Taught graduate-level course “Computer Networks”

**Ministry of National Defense** — Seoul, Korea
*Cyber Officer* — Aug 2023 - May 2025
- Deployed to Abu Dhabi, UAE for joint research on AI-based security (Apr 2024 - Mar 2025)

**Agency for Defense Development (ADD)** — Seoul, Korea
*Researcher, Cyber Technology Center* — Jul 2018 - Jul 2023
- Mentor: Changhee Choi

## Selected Publications

TIFS **I. Lee** *et al.*, “MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution”

TDSC **I. Lee** *et al.*, “UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design”

CAL **I. Lee** *et al.*, “LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis”

## Publications

### Preprints

(†: Corresponding Author)

P1 *Schema-Agnostic Knowledge Graph Construction via Hybrid Ontology Discovery for Cyber Threat Intelligence*
S. Kim, J. Kim, D. Kang, D. Kim, and **I. Lee**†
arXiv:2606.01208, May 2026.

## International Publications

(C: Conference, J: Journal)

C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, G. Ahn, S. Park, D. Bae, G. Kim, **I. Lee**, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (CCS'26), Nov. 2026.

C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (DAC'26), Jul. 2026.

J9 *MuCamp2: Generating Validated Cyber Campaign Variants via Constrained LLMs for Group Attribution*
K. Kim, J. Lee, **I. Lee**, I. Han, J. Lee, S. Shim, and C. Choi
IEEE Access, Vol. xx, pp. xx-xx, Jun. 2026.

J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*
**I. Lee**, D. Bae, S. Hong, and S. Lee
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.

J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.

J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*
**I. Lee**, K. Alteneiji, and M. Alghfeli
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.

J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*
**I. Lee** and C. Choi
IEEE Transactions on Information Forensics and Security (TIFS), Vol. 20, pp. 6162-6174, Jun. 2025.

J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*
**I. Lee** and W. Lee
IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.

J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*
**I. Lee** and C. Choi
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.

J2 *Exploiting TTP Co-occurence via GloVe-Based Embedding With ATT&CK Framework*
C. Shin, **I. Lee**, and C. Choi
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.

J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.

C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*
S. Shin, **I. Lee**, and C. Choi
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.

C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*
C. Choi, S. Shin, and **I. Lee**
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

## Domestic Publications (Korean)

(†: Corresponding Author)

D3 S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)

D2 H. Park and **I. Lee**†, "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.

D1 K. Kim and **I. Lee**†, "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

## Patents

- C. Choi and **I. Lee**, "Method for Augmentating Cyber Attack Campaign Data to Identify Attack Group, and

Security," Korea Patent Application Number. 10-2024-0176082, December 2, 2024.

- C. Choi, **I. Lee**, C. Shin, and S. Lee, "Information Identification Method and Electronic Apparatus Thereof," Korea Patent Application Number. 10-2024-0006106, January 15, 2024.
- C. Choi, C. Shin, S. Shin, S. Seo, and **I. Lee**, "Method for Training Attack Prediction Model and Device Therefor," U.S. Patent Application Number. 18/126,005; U.S. Patent Number. US20230308462A1, September 28, 2023.
- C. Choi, S. Shin, and **I. Lee**, “Apparatus, Method, Computer-readable Storage Medium and Computer Program for Generating Operation Code,” Korea Patent Application Number. 10-2019-0141865, November 07, 2019; Korea Patent Number. 10-2246797, April 30, 2021.

## Other Experience

AI Cyber Challenge (AIxCC), DARPA and ARPA-H, USA — Apr 2024 – Aug 2024

- Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities

SW Outsourcing Development, KCMVP-Certified Cryptographic Module — Jun 2017 – May 2018

- Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)

## Awards and Honors

- KU Graduate School Achievement Award — Feb 2026
  Korea University, Seoul, Republic of Korea
- Outstanding Paper Award — Nov 2025
  CISC-W’25, KIISC
- Certificate of Commendation (UAE-ROK Engagement Program) — Mar 2025
  United Arab Emirates Ministry of Defense
- Ambassador’s Commendation — Mar 2025
  Embassy of the Republic of Korea to the United Arab Emirates
- Full Tuition Scholarship (Korea University) — Mar 2014 – Feb 2018
  Ministry of National Defense, Republic of Korea

## Professional Service

**Reviewer**

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE/ACM Transactions on Networking (ToN), 2026
- IEEE Transaction on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

## Teaching Experience

- Lecturer, Computer Networks (SCS302, Fall 2025), Korea University — Sep 2025 - Dec 2025
- Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense — Sep 2024
- Instructor, Penetration Testing - Basic, UAE Ministry of Defense — Jul 2024

Jun 2026

## Mentoring Experience

- **Seonwoo Kim** (Major at Ministry of National Defense) Feb 2026 – Present
  Project: LLM-based knowledge graph construction for CTI [P1]
- **Sujin Park** (Ph.D. Student at Korea University) Jun 2025 – Present
  Project: Side-channel analysis for anomaly detection [D3]
- **Hyoungrok Kim** (M.S. Student at Korea University) Mar 2025 – Sep 2025
  Project: LLM-based TTP extraction for CTI [J7]
- **Hyunjun Park** (Navy Lieutenant at Ministry of National Defense) Nov 2024 – Feb 2025
  Project: DDoS detection via transfer learning [D2]
- **Kangmun Kim** (First Lieutenant at Cyber Operations Command) Jan 2024 – Sep 2024
  Project: Web shell detection via user behavior embedding [D1]

## Reference

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